

NFC batteryless e-ink tag

Recommended design: NT3H2211 (ISO 14443-A harvest + SRAM pass-through)\n+ STM32G071CBT6 + TPS61023 + GDEY042T81 4.2 in 400x300.\nPhone field powers the board. MCU streams the image, paints the panel, then the rail collapses.

Session: couple phone to the bezel antenna -> harvest charges VSTORE ->\nboost reaches 3.3 V -> supervisor releases PF2-NRST -> MCU boots ->\nSRAM mailbox drain over I2C -> enable EPD rail -> SSD1683 refresh -> deep sleep / field gone.

Do not hang bulk capacitance on NTAG VOUT. NXP limits that pin to 220 nF.\nCharge the tank through R1 (22 ohm) and a Schottky. Keep the phone on the coil for the whole refresh.

NFC front end



nfc.kicad_sch

Power



pwr.kicad_sch

MCU



mcu.kicad_sch

E-paper



epd.kicad_sch

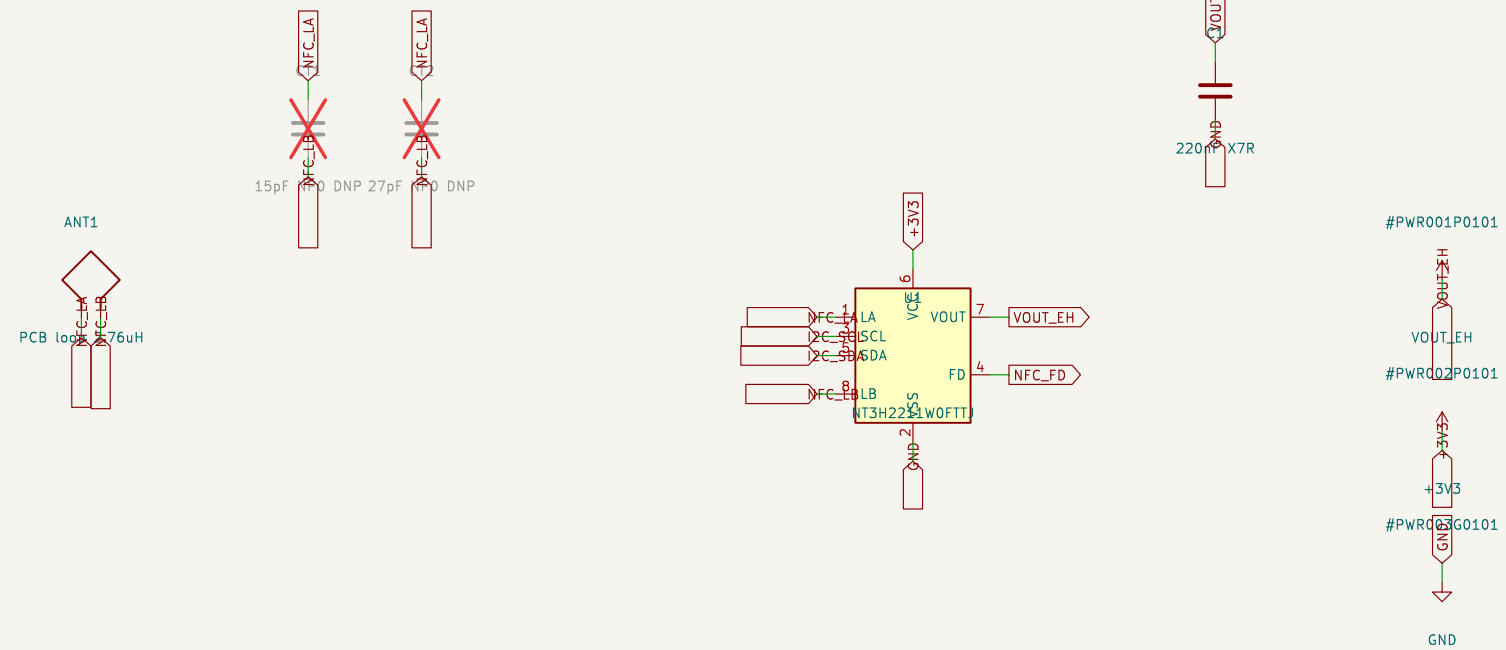
Board outline equals the GDEY042T81 module (91 mm x 77 mm).\nRoute a 2-turn Class-1 loop in the bezel; keep copper pour out of the antenna keepout.\nSee README.md for alternate designs (ST25DV, FM1280 module, 2.13 in, 5.83 in).

NFC batteryless e-ink tag		
playground		
Sheet: /		
File: nfc-eink.kicad_sch		
Title: NFC batteryless e-ink tag		
Size: A3	Date: 2026-09-04	Rev: 1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/5	

Antenna and NT3H2211W0FTTJ

Internal Cr = 50 pF. Target Lant = 2.76 uH at 13.56 MHz.
CT1/CT2 are NP0 tuning pads. Leave DNP until a VNA or phone-range sweep.

NXP max on **VOUT** is 220 nF. Do not add more here.



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KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

Id: 2/5

Harvest, tank, boost, EPD load, switch, reset

VOUT_EH -- D1 -- R1 -- VSTORE -- U2 boost --> +3V3

Hold off the EPD rail until VSTORE_DIV says the tank is ready.

$V_{out} = 0.595 \times (1 + 453k/100k) = 3.29\text{ V}$

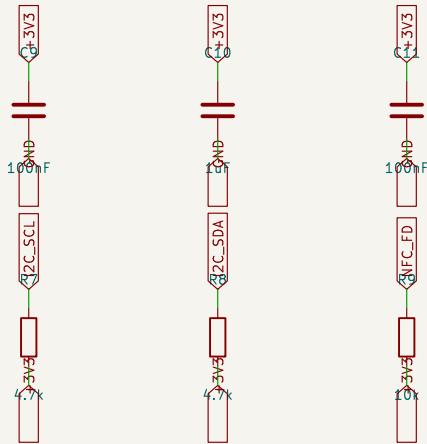
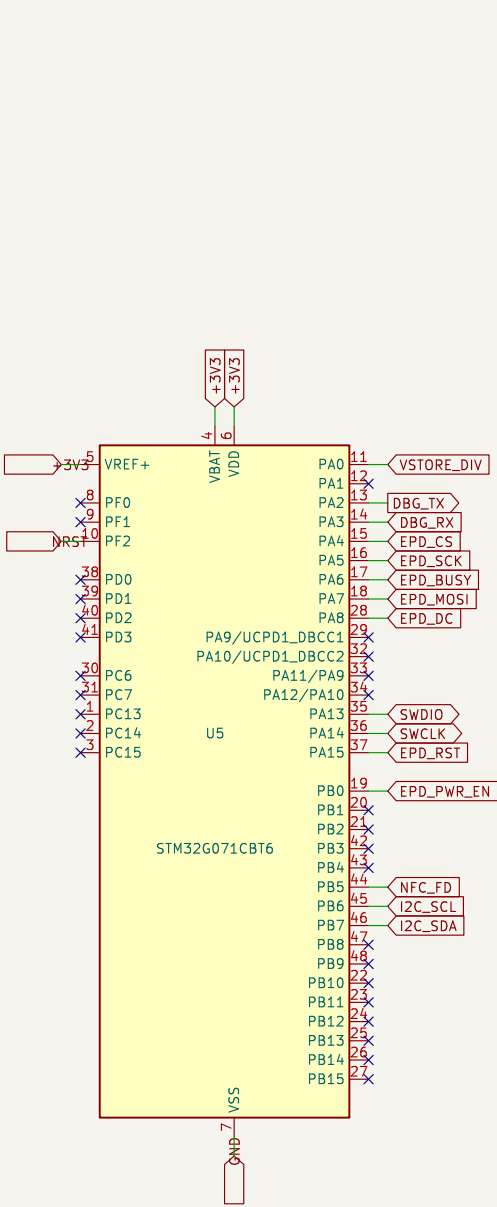
TCM809 is push-pull. No capacitor on NRST. R12 holds the pin low after VDD falls below 1 V.

R11 keeps TPS22917 OFF while PB0 is analog after reset.

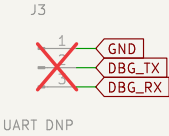
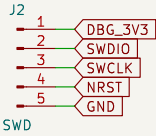
Close JP1 only when an ST-Link feeds J2 pin 1. Leave open for NFC power.

STM32G071CBT6 (HSI, no crystal)

36 KB RAM holds the 15 000-byte 4.2 in framebuffer.\nI2C1 on PB6/PB7. SPI1 on PA5/PA7. PF2 is nRESET on this package.

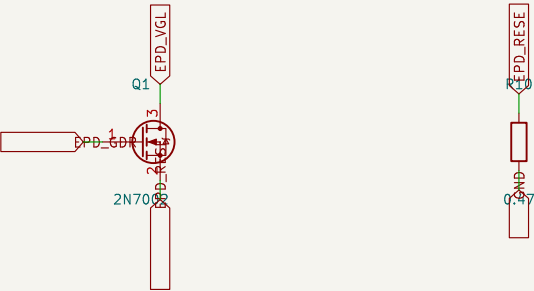
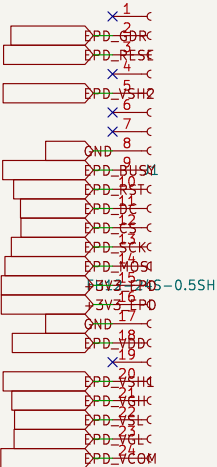
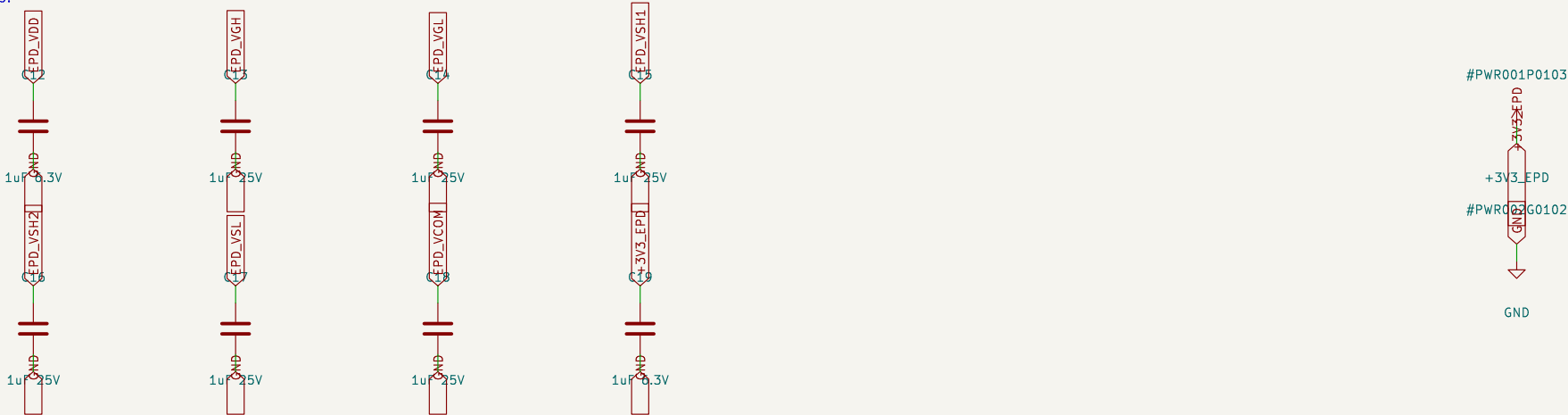


J2 1.27 mm: 3V3, SWDIO, SWCLK, NRST, GND



GDEY042T81 24-pin 0.5 mm FPC SSD1683

On-glass DC-DC still needs these host capacitors. Typical refresh is 13 mW / 5.6 mA at 3.0 V for 3 s.



SSD1683 booster sense MOSFET. Follow Good Display DESPI-C02 if a panel lot already has this on the FPC.